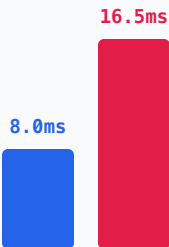


THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

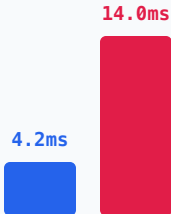
Deconstructing where time is lost: Nominal P50 (52.2 ms) vs. P99 Latency Tail Blowout (126.0 ms) Breaching World Deadlines

1. Transduction Photon / Exposure



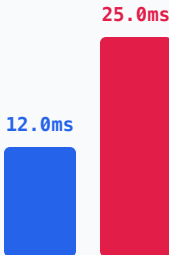
Rolling shutter
Integration time

2. DMA Transport MIPI CSI-2 Deserializer



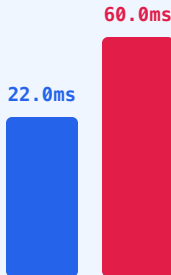
LPDDR5 memory bus
DMA buffer copies

3. Perception ViT Spatial Tokens



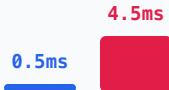
3D affordances
SE(3) projection

4. Policy Inference Diffusion / ACT Model



NPU compute stall
Weight bus contention

5. Inter-Core IPC Shared SRAM Mailbox



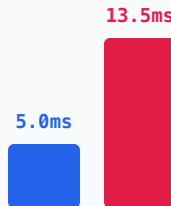
Seqlock copy
Lock-free mailbox

6. Safety Reflex 1000 Hz MCU (CBF QP)



QP solve < 65 μ s
Deterministic slack

7. Motor Torque L/R Rise Time



Inductive delay
Inverter PWM latch

■ P50 Nominal Execution

■ P99 Contention & Jitter Tail Blowout

----- Physical World Stability Deadline ($\tau_{\text{world}} = 800 \text{ ms}$)

P50 Total:  52.2 ms (Margin: +27.8 ms)

P99 Total:  126.0 ms (BREACH!)

$\tau_{\text{world}} = 800 \text{ ms}$

Systems Insight: While mean latency ($P50 = 52.2 \text{ ms}$) sits comfortably within stability margins, real-world memory crossbar contention and OS thread preemption inflate $P99$ to 126.0 ms, breaching the physical deadline and forcing open-loop drift unless arrested by a 1 kHz MCU safety reflex.